

IMO PHASE III LEVEL 2

BINOMIAL COEFFICIENTS(With Solution)

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1 Lucas' Theorem and Kummer's Theorem

What is the remainder of $\binom{m}{n} \pmod{p}$? And if $\binom{m}{n}$ is divisible by p , what is the p -adic valuation? These two questions are answered by **Lucas' Theorem** and **Kummer's Theorem** respectively.

Theorem 1.1 (Lucas' Theorem). Write the numbers n and k in base p : $n = \overline{n_d n_{d-1} \cdots n_0}_p$, $k = \overline{k_d k_{d-1} \cdots k_0}_p$. Then $\binom{n}{k} \equiv \prod_{i=0}^d \binom{n_i}{k_i} \pmod{p}$, with $\binom{n}{k} = 0$ when $n < k$. In particular, $\binom{n}{k}$ is divisible by p iff at least one of the base- p digits of n is greater than the corresponding base- p digit of k .

Lemma 1.2. For p prime and $x, r \in \mathbb{Z}$, $(1+x)^{p^r} \equiv 1 + x^{p^r} \pmod{p}$.

Proof. For all $1 \leq k \leq p-1$, $\binom{p}{k} \equiv 0 \pmod{p}$. Then we have

$$\begin{aligned} (1+x)^p &\equiv \binom{p}{0} + \binom{p}{1}x + \binom{p}{2}x^2 + \cdots + \binom{p}{p-1}x^{p-1} + \binom{p}{p}x^p \\ &\equiv 1 + x^p \pmod{p} \end{aligned}$$

Assume we have $(1+x)^{p^k} \equiv 1 + x^{p^k} \pmod{p}$. Then

$$\begin{aligned} (1+x)^{p^{k+1}} &\equiv \left((1+x)^{p^k}\right)^p \equiv \left(1 + x^{p^k}\right)^p \\ &\equiv 1 + x^{p^{k+1}} \pmod{p} \end{aligned}$$

□

Example 1.3 (Lucas). Suppose $p = 5, n = 482 = \overline{3412}_5, k = 176 = \overline{1201}_5$. By the lemma we have

$$\begin{aligned} (1+x)^{482} &\equiv (1+x)^{125(3)} \cdot (1+x)^{25(4)} \cdot (1+x)^{5(1)} + (1+x)^2 \pmod{5} \\ &\equiv (1+x^{125})^3 \cdot (1+x^{25})^4 \cdot (1+x^5)^1 \cdot (1+x)^2. \end{aligned}$$

Since the only way to get x^{176} is by three x^{125} terms, two x^{25} terms, zero x^5 terms, and one x^1 terms on the RHS. Hence the number of ways to get such term is

$$\binom{482}{176} \equiv \binom{3}{1} \binom{4}{2} \binom{1}{0} \binom{2}{1}.$$

Proof. (Lucas') The proof of Lucas' theorem follow a similar structure of the example. Another way to proof the theorem is by writting $n = Np + i, k = Kp + j$, and proceed by MI with some pascal's triangle properties. $\square$

Example 1.4 (lucas). Prove that the number of odd binomial coefficients in n -th row of Pascal triangle is equal to 2^r , where r is the number of 1's in the binary expansion of n .

Proof. Write $n = \overline{n_d n_{d-1} \cdots n_0}$, $\binom{n}{k}$ is odd iff $k_i \leq n_i \forall i$, and hence there are 2 choices for k_i if $n_i = 1$ and 1 choice otherwise, i.e. the number of odd entries equal 2^r . $\square$

Definition 1.5 (p -adic valuation). Given a non-zero integer n and a prime p , we define the p -adic valuation of n as the highest power of p that divides n , and write it as $v_p(n)$.

Theorem 1.6 (Kummer's Theorem). Given a prime p and integers m, n , $v_p\left(\binom{m}{n}\right)$ equals the number of carries in adding $m - n$ and n in base p .

Theorem 1.7 (Legendre's Formula). Let $n \in \mathbb{N}$, p be prime. Let $s_p(n)$ be the sum of the digits of n in base p . Then

$$v_p(n!) = \frac{n - s_p(n)}{p - 1}.$$

Proof. Let $n = \overline{a_k a_{k-1} \cdots a_0}_p$ be base p representation. We have $s_p(n) = \sum_{i=0}^k a_i$, then

$$\begin{aligned} v_p(n!) &= \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots \\ &= (a_k p^{k-1} + \cdots + a_1) + (a_k p^{k-2} + a_{k-1} p^{k-3} + \cdots + a_2) + \cdots + (a_k p + a_{k-1}) + a_k \\ &= a_k \frac{p^k - 1}{p - 1} + a_{k-1} \frac{p^{k-1} - 1}{p - 1} + \cdots + a_1 \\ &= \frac{1}{p - 1} ((a_k p^k + a_{k-1} p^{k-1} + \cdots + a_0) - (a_k + a_{k-1} + \cdots + a_0)) \\ &= \frac{n - s_p(n)}{p - 1}. \end{aligned} \quad \square$$

Proof. (Kummer's) Write $m = \overline{a_r a_{r-1} \cdots a_0}_p, m - n = \overline{b_r b_{r-1} \cdots b_0}_p, n = \overline{c_r c_{r-1} \cdots c_0}_p$. Define carry func γ :

$$\gamma_i = \begin{cases} 1 & \text{if } b_i + c_i + \gamma_{i-1} \geq p, \\ 0 & \text{if } b_i + c_i + \gamma_{i-1} < p. \end{cases} \quad \forall 0 \leq i < r, \gamma_{-1} = 0.$$

Since a_r is the leading coeff of m , $\gamma_r = 0$. And observe that $a_i = b_i + c_i + \gamma_{i-1} - p\gamma_i \forall 0 \leq i \leq r$. By Legendre,

$$\begin{aligned} v_p\left(\binom{m}{n}\right) &= v_p(m!) - v_p((m - n)!) - v_p(n!) \\ &= \frac{m - s_p(m)}{p - 1} - \frac{m - n - s_p(m - n)}{p - 1} - \frac{n - s_p(n)}{p - 1} \\ &= \frac{\sum_{i=0}^r b_i + c_i - a_i}{p - 1} \\ &= \frac{p\gamma_0 + (p\gamma_1 - \gamma_0) + \cdots + (p\gamma_{r-1} - \gamma_{r-2}) - \gamma_{r-1}}{p - 1} \\ &= \gamma_0 + \gamma_1 + \cdots + \gamma_{r-1}. \end{aligned} \quad \square$$

2 Some other Theorem

Theorem 2.1 (Wilson's Theorem). For any prime p (and for primes only), $(p-1)! \equiv -1 \pmod{p}$.

Theorem 2.2 (Wolstenholme's Theorem). If $p \geq 5$, then $\binom{2p}{p} \equiv 2 \pmod{p^3}$, or equivalently, $\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}$.

Proof. We have $\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2$. Since $\binom{p}{0} = \binom{p}{p} = 1$, we need to prove $\sum_{k=1}^{p-1} \binom{p}{k}^2 \equiv 0 \pmod{p^3}$. Since all the binomial coefficients are divisible by p , we can rewrite it as

$$\sum_{k=1}^{p-1} \left(\frac{1}{p} \binom{p}{k} \right)^2 \equiv 0 \pmod{p}.$$

But we have

$$\frac{1}{p} \binom{p}{k} = \frac{(p-1) \cdots (p-(k-1))}{k(k-1) \cdots 1} \equiv \frac{1(-1) \cdots (-(k-1))}{k(k-1) \cdots 1} \equiv \pm \frac{1}{k} \pmod{p},$$

thus

$$\sum_{k=1}^{p-1} \left(\frac{1}{p} \binom{p}{k} \right)^2 \equiv \sum_{k=1}^{p-1} \left(\pm \frac{1}{k} \right)^2 \equiv \sum_{k=1}^{p-1} \left(\frac{1}{k} \right)^2 = \sum_{k=1}^{p-1} k^2 = \frac{p(p-1)(2p-1)}{6} \equiv 0 \pmod{p}$$

as when $1/k$ go through 1 to $p-1$, so does k . □

Remark. An equivalent formulation is the congruence $\binom{ap}{bp} \equiv \binom{a}{b} \pmod{p^3}$ for $p \geq 5$, which we will not cover the proof here.

3 Problems for Discussion

Problem 1. For prime p , prove that $\binom{p}{n} - \lfloor \frac{n}{p} \rfloor$ is divisible by p .

Ans. Let $n = a_0 + a_1p + a_2p^2 + a_3p^3 + \cdots$ for some integers $a_0, a_1, a_2, \dots$ where $0 \leq a_i \leq p-1$ for all i . Then, by Lucas' Theorem,

$$\begin{aligned} \binom{n}{p} &\equiv \binom{a_0}{0} \binom{a_1}{1} \binom{a_2}{0} \binom{a_3}{0} \cdots \equiv \binom{a_1}{1} \\ &\equiv a_1 \equiv a_1 + a_2p + a_3p^2 + \cdots \\ &\equiv \left\lfloor \frac{n}{p} \right\rfloor \pmod{p} \end{aligned}$$

Problem 2. Prove that

- All the binomial coefficients $\binom{n}{k}$, where $0 < k < n$, are divisible by p iff n is a power of p .
- All the binomial coefficients $\binom{n}{k}$, where $0 \leq k \leq n$ are not divisible by p iff $n+1$ is divisible by p^d , in other words, all the digits of n , except the leftmost, in base p are equal to $p-1$.

Ans.

1. If p does not divide $\binom{n}{k}$, then there are no carries when we add k and $n-k$ in base p . For a fixed n it means that we can choose i -th digit of k in base p by $n_i + 1$ ways. Hence we have a row with 2 elements only not divisible by p .
2. If $p^d \mid (n+1)$, then $n = \overline{a(p-1)(p-1) \cdots (p-1)}$ in base p . Then for any k , $0 \leq k \leq n$, each digit of k does not exceed the corresponding digit of n . Therefore all the binomial coefficients $\binom{n}{k_i}$ are not equal to 0 and $\not\equiv 0 \pmod{p}$. By Lucas' theorem $\binom{n}{k}$ is not divisible by p .

The reverse statement. Assume that all the coefficients $\binom{n}{k}$ are not divisible by p , but n is not the number of the form $\overline{a(p-1)(p-1) \cdots (p-1)}$. Therefore one of its digits, say, n_i is less than $p-1$. Choose $k = (p-1) \cdot p^i$. Then $k_i = p-1$ and hence $\binom{n_i}{k_i} = 0$, and $p \mid \binom{n}{k}$ by Lucas' theorem. A contradiction.

Problem 3. (VN TST 2010 P6) Let S_n be sum of squares of the coefficient of the polynomial $(1+x)^n$. Prove that $S_{2n} + 1$ is not divisible by 3.

Ans. $S_n = \sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$ (by double counting). Then $S_{2n} + 1 = \binom{4n}{2n} + 1$.

Lemma : If $\binom{2x}{x}$ is not divisible by p . and $x = (a_0, \dots, a_t)_p$ then $0 \leq a_i \leq \frac{p-1}{2}$ and $2x = (2a_0, \dots, 2a_t)_p$

Proof: obvious by either Lucas or Kummer.

So by the lemma $2n = (a_0, \dots, a_t)_3$ s.t $a_i \in \{0, 1\}$ So FSoC $-1 = \binom{4n}{2n} = \sum_{i=0}^t \binom{2a_i}{a_i} \pmod{3}$ Notice each term of RHS becomes -1 if $a_i = 1$ and otherwise 1 . so we should have odd number of them equal to 1 . But $2n = a_0 + a_1 3 + \dots + a_t 3^t$ ($\spadesuit$) odd number of a_i 's are 1 so RHS of ($\spadesuit$) is odd but LHS of it is even. Contradiction ■

Problem 4. Prove that $\text{lcm}(1, 2, \dots, n) = \text{lcm}(\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n})$ iff $n+1$ is prime.

Ans. Lemma: if $n \in N$ then we have: $\frac{1}{n+1}(\text{lcm}(1, 2, \dots, n+1)) = \text{lcm}(\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n})$

Proof: Let $p^r \leq n < p^{r+1}$, we have $\text{lcm}(\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}) \mid \frac{1}{n+1}(\text{lcm}(1, 2, \dots, n+1))$ as $v_p(\binom{n}{k}) \leq r = \frac{1}{n+1} \text{lcm}(1, 2, \dots, n+1)$ by Kummer's. And put $k = p^r - 1$, we have $p^r \mid \binom{n}{k}$ by Kummer's. Hence the equality.

Problem 5. (Olimpiada Rioplatense 2015 Level 3 P5) For a positive integer number n we denote $d(n)$ as the greatest common divisor of the binomial coefficients $\binom{n+1}{n}, \binom{n+2}{n}, \dots, \binom{2n}{n}$. Find all possible values of $d(n)$

Ans. We see that for any $p \mid d(n)$ then $p \mid n+1$. So pick a prime divisor p of $d(n)$ then $p \mid n+1$. We can write n base p as $n = \overline{a_k a_{k-1} \dots a_1 (p-1)}_p$ where $0 \leq a_k, a_{k-1}, \dots, a_1 \leq p-1$ and $a_k \neq 0$.

If there exists a number $a_i \leq p-2$ then we have $n < n + p^i \leq n + p^k < 2n$ and

$$n + p^i = \overline{a_k a_{k-1} \dots a_{i+1} (a_i + 1) a_{i-1} \dots a_1 (p-1)}_p.$$

Therefore, according to Lucas theorem, we have

$$\binom{n+p^i}{n} \equiv \binom{a_k}{a_k} \dots \binom{a_{i+1}}{a_{i+1}} \binom{a_i+1}{a_i} \dots \binom{a_1}{a_1} \equiv a_i + 1 \pmod{p}.$$

Hence such existence of $a_i \leq p-2$ implies $p \nmid \binom{n+p^i}{n}$, a contradiction. Thus, we must have $a_i = p-1$ for all $1 \leq i \leq k$ or $n = p^{k+1} - 1$ for $k \geq 0$. This representation of n is unique, so this means that if $n \neq p^l - 1$ for some prime p and $l \geq 1$ then $d(n) = 1$.

If $n = p^l - 1$, we will prove that $d(n) = p$. Indeed, since there will be no other prime $p_1 \neq p$ and $l_1 \geq 1$ so that $n = p^l - 1 = p_1^{l_1} - 1$ so $d(n) = p^x$. Now, by using Kummer's theorem for $n = p^l - 1$, we will have $\nu_p \left(\binom{p^{l-1} + n}{n} \right) = 1$ since there is one carry when we plus $n = p^l - 1$ and p^{l-1} in base p . Therefore, $\nu_p(d(n)) = 1$. Thus, $d(n) = p$.

In conclusion, if $n \neq p^l - 1$ for some prime p and positive integer l then $d(n) = 1$. If $n = p^l - 1$ then $d(n) = p$.

Problem 6. Prove that if we consider all the elements of the two sets

$$\left\{ \binom{2^n-1}{1}, \binom{2^n-1}{3}, \binom{2^n-1}{5}, \dots, \binom{2^n-1}{2^n-1} \right\} \quad \text{and} \quad \{1, 3, 5, \dots, 2^n-1\}$$

as a reminders modulo 2^n , then these sets coincide.

Ans.

All the binomial coefficients in the problem statement are odd by Lukas' theorem, therefore, it is sufficient to check that all the numbers $\binom{2^n-1}{1}, \binom{2^n-1}{3}, \dots, \binom{2^n-1}{2^n-1}$ have distinct reminders modulo 2^n .

Assume by the contrary that $\binom{2^n-1}{k} \equiv \binom{2^n-1}{m} \pmod{2^n}$ for odd k and m , $k > m$. Observe that

$$\begin{aligned} \binom{2^n-1}{k} &= \binom{2^n}{k} - \binom{2^n-1}{k-1} = \binom{2^n}{k} - \binom{2^n}{k-1} + \binom{2^n-1}{k-2} = \dots = \\ &= \binom{2^n}{k} - \binom{2^n}{k-1} + \binom{2^n}{k-2} - \dots - \binom{2^n}{m+1} + \binom{2^n-1}{m}. \end{aligned}$$

In particular

$$\binom{2^n}{k} - \binom{2^n}{k-1} + \binom{2^n}{k-2} - \dots - \binom{2^n}{m+1} \equiv 0 \pmod{2^n}.$$

Calculate the exponent $v_2\left(\binom{2^n}{r}\right)$ by Kummer's theorem. If $v_2(r) = a$ then we have $n - a$ carries in addition r and $2^n - r$ (it is clear by the standard algorithm of addition), hence $v_2\left(\binom{2^n}{r}\right) = n - a$. In particular $2^n \mid \binom{2^n}{r}$ for odd r , that allows us to consider only one half of summands:

$$\binom{2^n}{k-1} + \binom{2^n}{k-3} + \dots + \binom{2^n}{m+1} \equiv 0 \pmod{2^n}.$$

Now all the $\binom{2^n}{i}$ in the left hand side have even parameter i , therefore $v_2\left(\binom{2^n}{x}\right) < n$.

We will prove that this congruence is impossible and obtain a contradiction. Choose x with minimal $v_2\left(\binom{2^n}{x}\right)$. Since $v_2\left(\binom{2^n}{x}\right) < n$ and the whole sum is divisible by 2^n , there exists y , for which $v_2\left(\binom{2^n}{x}\right) = v_2\left(\binom{2^n}{y}\right)$. Then the binary representations of x and y end with equal number of 0's, and hence there exists z between x and y which binary representation ends with bigger number of 0's. Then $v_2\left(\binom{2^n}{z}\right) < v_2\left(\binom{2^n}{x}\right)$, a contradiction.

Problem 7. (Iran TST 2012 -first day- problem 1) Find all positive integers $n \geq 2$ such that for all integers i, j that $0 \leq i, j \leq n$, $i + j$ and $\binom{n}{i} + \binom{n}{j}$ have same parity.

Ans. Let $i = 0$, for all $0 \leq j = 2r \leq n$, $\binom{n}{2r} + 1$ is even, so $\binom{n}{2r}$ is odd. Denote $n = (\overline{n_t n_{t-1} \dots n_1 n_0})_2$, $2r = (\overline{k_t k_{t-1} \dots k_1 0})_2$, $n_t = 1$. By Lucas theorem $\prod_{i=0}^t \binom{n_i}{k_i} \equiv \binom{n}{2r} \equiv 1 \pmod{2}$, so $\binom{n_i}{k_i} = 1$ for all $0 \leq i \leq t$, so $n_0 = 0$. If there exist integer $1 \leq s \leq t-1$ satisfies $n_s = 0$, we take $2r = 2^s$, then $\binom{n_s}{k_s} = \binom{0}{1} = 0$, contradiction. So $n = (\overline{11 \dots 10})_2 = 2^{t+1} - 2$.

On the other hand if $n = (\overline{11 \dots 10})_2$, because $C_1^0 = C_1^1 = C_0^0 = 1$, for any $i = (\overline{a_t a_{t-1} \dots a_1 a_0})_2$, by Lucas theorem $\binom{n}{i} \equiv \prod_{i=0}^t \binom{n_i}{a_i} \equiv \binom{0}{a_i} \equiv a_0 - 1 \equiv i - 1 \pmod{2}$, similarly we have $\binom{n}{j} \equiv j - 1 \pmod{2}$, so $\binom{n}{i} + \binom{n}{j} \equiv i + j \pmod{2}$. Above all, $n = 2^{t+1} - 2$, $t \in \mathbb{Z}^+$.

Problem 8. Show that, for every prime number p and all integer r , there is a non negative integer n , such that:

$$\binom{2n}{n} \equiv r \pmod{p}$$

Ans. Claim 1. If there exists n_0 so $\binom{2n_0}{n_0} \equiv r \pmod{p}$ then there exists n so $\binom{2n}{n} \equiv 2r \pmod{p}$.

Proof. Pick $n = p^k + n_0$ with $p^k > n_0$ then according to Lucas' theorem we have

$$\binom{2p^k + 2n_0}{p^k + n_0} \equiv \binom{2}{1} \binom{2n_0}{n_0} \equiv 2r \pmod{p}.$$

Claim 2. If there exists n_0 so that $\binom{2n_0}{n_0} \equiv r \pmod{p}$ with $r < p/2$ then there exists n so $\binom{2n}{n} \equiv 2r - 1 \pmod{p}$.

Proof. Note that we have

$$\begin{aligned} 2^{p-2} \binom{2r-2}{r-1}^{p-2} \binom{2r}{r} &= 2^{p-1} (2r-1) r^{p-2} \left(\frac{(r+1) \cdots (2r-2)}{(r-1)!} \right)^{p-1}, \\ &\equiv (2r-1) r^{p-2} \pmod{p}. \end{aligned}$$

Hence, if we pick $n = n_0 \cdot p^k + (\underbrace{1 \cdots 1}_{p-2} \underbrace{(r-1) \cdots (r-1)}_{p-2})_p$ for large enough k then according to Lucas' theorem,

we have

$$\binom{2n}{n} \equiv \binom{2n_0}{n_0} \binom{2}{1}^{p-2} \binom{2r-2}{r-1}^{p-2} \binom{2r}{r} \equiv r^{p-1} (2r-1) \equiv 2r-1 \pmod{p}.$$

From these two claims, by induction, starting with $\binom{2}{1} \equiv 2 \pmod{p}$, we can follow that if there exists n so $\binom{2n}{n} \equiv r \pmod{p}$ for all $r \leq k$ then there also exists n so $\binom{2n}{n} \equiv k+1 \pmod{p}$. And $\binom{p+1}{(p+1)/2} \equiv 0 \pmod{p}$. We are done.

4 Problems (Combinatorial)

Problem 1. (2019 Pan-African Mathematics Olympiad P6) Find the 2019th strictly positive integer n such that $\binom{2n}{n}$ is not divisible by 5.

Ans. Kummer's theorem tells us that that number of times a prime p divides $\binom{n}{m}$ is equal to the number of carries when calculating $n = m + (n - m)$ in base p . In this case, we see that there must be no carries when calculating $n + n = 2n$ in base 5, and so the base 5 digits of n must all be at most 2.

Lucas' Theorem implies that $\binom{2n}{n}$ is divisible by 5 if any of the base 5 digits of $2n$ are smaller than the corresponding base 5 digit of n . This is again equivalent to any of the digits of n being larger than 2.

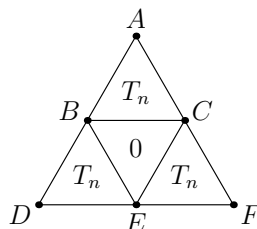
We are thus seeking the 2019th natural number n such that the base 5 expansion of n consists only of the digits 0, 1, and 2. If we ignore the fact that the expansion is a number in base 5, then the string of digits can also be interpreted as a number in base 3, and so we see that the 2019th valid string of digits is given by the base 3 expansion of 2019, which is 2202210_3 . It follows that the number that we are looking for is $2202210_5 = 37805$.

Problem 2. Wanda the Worm likes to eat Pascal's triangle. One day, she starts at the top of the triangle and eats $\binom{0}{0} = 1$. Each move, she travels to an adjacent positive integer and eats it, but she can never return to a spot that she has previously eaten. If Wanda can never eat numbers a, b, c such that $a + b = c$, prove that it is possible for her to eat 100,000 numbers in the first 2011 rows given that she is not restricted to traveling only in the first 2011 rows.

(Here, the $n + 1$ st row of Pascal's triangle consists of entries of the form $\binom{n}{k}$ for integers $0 \leq k \leq n$. Thus, the entry $\binom{n}{k}$ is considered adjacent to the entries $\binom{n-1}{k-1}$, $\binom{n-1}{k}$, $\binom{n}{k-1}$, $\binom{n}{k+1}$, $\binom{n+1}{k}$, $\binom{n+1}{k+1}$.)

Ans. Lemma: Wanda can eat all the odd entries from rows 0 to $2^n - 1$ and ending at $\binom{2^n-1}{0}$ or $\binom{2^n-1}{2^n-1}$.

Proof of Lemma: The key idea is that Pascal's triangle mod 2 up to row $2^{n+1} - 1$ (call it T_{n+1}) can be viewed as three copies of the triangle till row $2^n - 1$, or T_n , in the following fashion:



The statement is obvious for T_1 . Suppose the problem is true for T_n . Use the inductive hypothesis to get all the odds in ABC , and end at B . Now use the inductive hypothesis on BDE to get to E . Use the inductive hypothesis on ECF to get to F , and this finishes (note that T_n is symmetric under rotation by $2\pi/3$). This completes the proof. ■

Now, note that there are no odd a, b, c with $a + b = c$. Note that T_n has 3^n odd entries (proof is by induction). We see that using our strategy in the lemma, Wanda can eat all the odd entries till row 2047. Above row 2011 are at least a T_{10} , two T_9 s, and two T_8 s, for a total of at least

$$3^{10} + 2 \cdot 3^9 + 2 \cdot 3^8 = 111537 \geq 100,000$$

odd numbers, as desired.

Problem 3. $n(n \geq 3)$ integers $a_1, a_2, \dots, a_n$ lie around a circle in that order. In every operation, we calculate the absolute value of difference between each pair of two adjacent numbers to get n new integers, and then set them around the circle without disorganizing the order. Determine all the positive integer n , such that for any n integers, we can take finite operations to make all n integers are the same eventually.

Ans. The answer is $n = 2^k$.

To see this, it's helpful to first consider an example, say $n = 4$. After the first step, all the integers are nonnegative. From this point on, the largest number cannot increase. However, considering the various cases

for the integers modulo 2, it becomes apparent that no matter the initial values, after at most 4 steps all the integers become even. At this point we may as well factor a two out and repeat the previous argument over and over again, until all values eventually become zero. It turns out modulo 2 considerations are useful in the other cases for n , too.

If n is odd, it's fairly clear that all integers can never end up even being the same mod 2 (unless they started that way) - the integers before the final step must necessarily be $a_1, a_2, \dots, a_n = 1, 1, \dots, 1$, which means before that, $a_1, a_3, a_5, \dots, a_n$ had the same parity, yet a_1 and a_n had different parities, which is a contradiction.

Note, however, that if n is not a solution, $2n$ can't be, either. To see this, consider a configuration which shows n isn't a solution, namely one which eventually cycles modulo 2. Merely concatenating two copies of this configuration trivially also leads to a set of $2n$ integers which cycles modulo 2 in the exact same manner - the two halves essentially operate independently. For instance, using $n = 3$, $101 \rightarrow 110 \rightarrow 011 \rightarrow 101$ and $101101 \rightarrow 110110 \rightarrow 011011 \rightarrow 101101$. Hence, the only potential solutions remaining are $n = 2^k$.

Finally, when $n = 2^k$, first consider a configuration with only one 1, namely $a_1, a_2, \dots, a_n = 1, 0, \dots, 0$. After the first step, it becomes $1, 1, 0, \dots, 0$, then $1, 0, 1, 0, \dots, 0$, then $1, 1, 1, 1, 0, \dots, 0$, then $1, 0, 0, 0, 1, 0, \dots, 0$ and so on. It's clear that, at least for the first few (actually, $n - 1$) steps, the numbers are equivalent to the i 'th row of the Pascal triangle. However, by Lucas' theorem, on the $n - 1$ 'st step, all entries of the Pascal triangle will be odd - hence on the n 'th step, all numbers must become congruent to 0 mod 2, at which point we may finish the proof as we did in the example in the beginning.

Of course, if the numbers all reach 0 mod 2 for each configuration with only one 1 (as above), we can use a superposition argument to argue that, indeed, the numbers must necessarily do so for any configuration, which concludes the proof.

Problem 4. (2019 China TST Test 4 P6) Given positive integer n, k such that $2 \leq n < 2^k$. Prove that there exist a subset A of $\{0, 1, \dots, n\}$ such that for any $x \neq y \in A$, $\binom{y}{x}$ is even, and

$$|A| \geq \frac{\binom{k}{\lfloor \frac{k}{2} \rfloor}}{2^k} \cdot (n + 1)$$

Ans. Start by writing $n + 1$ in binary, so $n + 1 = 2^{a_1} + \dots + 2^{a_m}$ for $a_1 > \dots > a_m$. We'll use the example $n = 154$, where $n + 1$ in binary is 10011011. For each 1 in the binary representation, create a prefix by replacing that 1 with a 0 and keeping everything before that the same. Construct values q_i according to the rule

$$q_i = \min(q_{i-1} - 1, \lceil a_i/2 \rceil)$$

and the initial value is $q_1 = \lceil a_1/2 \rceil$. For our example, we have the following values:

prefix	a_i	q_i
0...	7	4
1000...	4	2
10010...	3	1
1001100...	1	0
10011010	0	-1

For each i , let S_i consist of all numbers whose binary representation consists of the i th prefix in the table followed by a string of a_i digits, of which exactly q_i are 1. If q_i is negative, then S_i is empty.

We claim $A_n = \bigcup S_i$ is a valid subset. By Lucas' theorem, this is equivalent to no binary representation of one number covering the binary representation of another. By the prefixes, if $y \in S_j$ and $x \in S_i$ are elements of A in binary form, then y can cover x only if $j \geq i$. By construction, the number of 1s of elements of S_j is at most the number of 1s of elements of S_i , so no covering occurs, as desired.

Let $f(n)$ denote the size of the set A_n obtained by this construction. It remains to show the lower bound

$$f(n) = \sum_{i=1}^m \binom{a_i}{q_i} \geq \frac{\binom{k}{\lfloor k/2 \rfloor}}{2^k} (n + 1).$$

We first resolve the case where $q_i = q_{i-1} - 1$ always. The case $n + 1 = 2^{a_1}$ is special, and equality holds for $k = a_1$. From now on, let $k = a_1 + 1$. We wish to show that

$$\frac{f(n)}{n + 1} = \frac{\sum_{i=1}^m \binom{a_i}{q_i}}{\sum_{i=1}^m 2^{a_i}} \geq \frac{\binom{k}{\lfloor k/2 \rfloor}}{2^k}.$$

[The goal is to reduce to the case $a_i = a_{i-1} - 1$, but I made a mistake here. I need to fix this.]

Then

$$f(n) = \sum_{i=1}^m \binom{a_1 + 1 - i}{q_1 + 1 - i} = \sum_{i=1}^m \binom{a_1 + 1 - i}{\lfloor a_i/2 \rfloor} = \binom{a_1 + 1}{\lfloor a_1/2 \rfloor + 1} - \binom{a_1 + 1 - m}{\lfloor a_1/2 \rfloor + 1}.$$

Meanwhile,

$$n + 1 = \sum_{i=1}^m 2^{a_1 + 1 - i}.$$

Now we consider the effect of increasing m by 1. Then $f(n)$ increases by $\binom{a_1 - m}{\lfloor a_1/2 \rfloor}$ while $n + 1$ increases by $2^{a_1 - m}$. Since

$$\frac{\binom{a_1 - m}{\lfloor a_1/2 \rfloor}}{2^{a_1 - m}} \leq \frac{\binom{a_1}{\lfloor a_1/2 \rfloor}}{2^{a_1}} \leq \frac{\binom{k}{\lfloor k/2 \rfloor}}{2^k},$$

we can assume the worst case scenario of $m = a_1 + 1$. In this case we have

$$\frac{f(n)}{n + 1} = \frac{\binom{a_1 + 1}{\lfloor a_1/2 \rfloor + 1}}{2^{a_1 + 1} - 1} > \frac{\binom{a_1 + 1}{\lfloor a_1/2 \rfloor + 1}}{2^{a_1 + 1}} = \frac{\binom{k}{\lfloor k/2 \rfloor}}{2^k}.$$

We now deal with the remaining cases by strong induction. Suppose for some $i > 1$, we have $q_i = \lceil a_i/2 \rceil$. Let $\ell + 1 = 2^{a_i} + 2^{a_{i+1}} + \dots + 2^{a_m}$. Then the construction from step i onward is the same as the construction for ℓ , just with everything shifted, i.e.

$$A_n = A_{n-\ell-1} \sqcup ((n-\ell) + A_\ell).$$

Therefore $f(n) = f(n - \ell - 1) + f(\ell)$. By the inductive hypothesis, if $n < 2^k$ then

$$f(n - \ell - 1) + f(\ell) \geq \frac{\binom{k}{\lfloor k/2 \rfloor}}{2^k} (n - \ell + \ell + 1) = \frac{\binom{k}{\lfloor k/2 \rfloor}}{2^k} (n + 1)$$

as desired.

5 Problems (Functional Equation)

Problem 1. Determine all functions f defined on the set of all positive integers and taking non-negative integer values, satisfying the three conditions:

- (i) $f(n) \neq 0$ for at least one n ;
- (ii) $f(xy) = f(x) + f(y)$ for every positive integers x and y ;
- (iii) there are infinitely many positive integers n such that $f(k) = f(n - k)$ for all $k < n$.

Ans. The only functions that work are $f(x) = f(p) \cdot \nu_p(x)$ for some prime p with $f(p) > 0$

By the first condition $\exists p$ such that $f(p) > 0$. Call all n satisfying (iii) as "good". For a good n we have :

$$f\left(\binom{n-1}{k}\right) = \sum_{i=1}^{n-1} f(i) - \sum_{i=1}^k f(i) - \sum_{i=1}^{n-k-1} f(i) = \sum_{i=n-k}^{n-1} f(i) - \sum_{i=1}^k f(i) = 0$$

Now note that if $p \mid m$, we must have that $f(m) > 0$. Hence for all large enough good n , we have that $p \nmid \binom{n-1}{k}$. By Lucas's theorem we have that $n = a \cdot p^b$ for some $a < p$.

Now assume there exists a prime $p' \neq p$ with $f(p') > 0$. Then all large enough good n must also be of the form $a' \cdot p'^{b'}$. However only finitely many such n can exist, so p is the only prime satisfying $f(p) > 0$.

Now its easy to that $f(x) = f(p) \cdot \nu_p(x)$ for all $x \in \mathbb{N}$.

Problem 2. Show that a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ of plus minus one is periodic of period a power of 2 if and only if $\varepsilon_n = (-1)^{P(n)}$ where $P(n)$ is a polynomial with rational coefficients which is integer-valued.

Ans. First, consider that $\varepsilon_n = (-1)^{P(n)}$, where $P(n)$. Let N be a positive integer, such that $Q(x) = N \cdot P(x) \in \mathbb{Z}[x]$. Put an integer $k > \nu_2(N)$, then $Q(n + 2^k) \equiv Q(n) \pmod{2^k}$. Since $k > \nu_2(N)$, dividing N from both sides yields $P(n) \equiv P(n + 2^k) \pmod{2}$, i.e. $(\varepsilon_n)_{n \in \mathbb{N}}$ is periodic of period 2^k .

Next, suppose $(\varepsilon_n)_{n \in \mathbb{N}}$ is a periodic sequence of period 2^k , our aim is to construct such $P(n)$. Let

$$p(x) = \binom{x}{2^k - 1} = \frac{x(x-1) \cdots (x-2^k+2)}{(2^k-1)!}$$

clearly $p(x)$ is a polynomial with rational coefficients which is integer-valued. By Lucas' Theorem, $p(m)$ is odd iff $m \equiv -1 \pmod{2^k}$, $p(m)$.

Let $1 \leq x_1, \dots, x_t < 2^k$ be all the index i such that $\varepsilon_i = -1$. Then $P(n) = \sum_{j=1}^t p(n-1-x_j)$. We have $P(n)$ takes odd values if and only if $n \equiv x_j \pmod{2^k}$ for some $1 \leq j \leq t$. That is to say, $\varepsilon_n = (-1)^{P(n)}$.

Problem 3. (2009 USA TST P3) For each positive integer n , let $c(n)$ be the largest real number such that

$$c(n) \leq \left| \frac{f(a) - f(b)}{a - b} \right|$$

for all triples (f, a, b) such that

- f is a polynomial of degree n taking integers to integers, and
- a, b are integers with $f(a) \neq f(b)$.

Find $c(n)$.

Ans. Let $L = \text{lcm}(1, \dots, n)$. We claim the answer is $c(n) = 1/L$.

By shifting suitably, we seek

$$c(n) = \min_{\substack{f(0)=0 \\ \deg f=n}} \min_{\substack{x \\ f(x) \neq 0}} \left| \frac{f(x)}{x} \right|.$$

Taking the basis $\binom{x}{1}, \binom{x}{2}, \dots, \binom{x}{n}$ for $\mathbb{Z} \rightarrow \mathbb{Z}$ polynomials, we may write $f(x) = a_1 \binom{x}{1} + a_2 \binom{x}{2} + \dots + a_n \binom{x}{n}$. Now for a fixed x the smallest nonzero value of $f(x)$ across all choices of f is exactly $\gcd_{1 \leq k \leq n} \binom{x}{k}$, by using Euclidean algorithm on a_k . So we now interchange the order of the min operators to obtain

$$c(n) = \min_{x \neq 0} \frac{|G(x)|}{x} \quad \text{where} \quad G(x) \stackrel{\text{def}}{=} \gcd \left(\binom{x}{1}, \binom{x}{2}, \dots, \binom{x}{n} \right).$$

First, we note $G(L) = 1$, since if p^e exactly divides L (even if $e = 0$) then $p \nmid \binom{L}{p^e}$ by Lucas theorem, say. Thus $c(n) \leq 1/L$.

On the other hand since

$$L \cdot \binom{x}{k} = L \cdot \frac{x}{k} \binom{x-1}{k-1} = x \cdot \frac{L}{k} \binom{x-1}{k-1} \in x \cdot \mathbb{Z}$$

we see that $L \cdot G(x) \geq x$ for any $x \in \mathbb{Z}$. Hence $c(n) \geq 1/L$, as desired.

Problem 4. (2016 Taiwan 1st TST IMO Mock P3) Let $\mathbb{Z}^+$ denote the set of all positive integers. Find all surjective functions $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ that satisfy all of the following conditions: for all $a, b, c \in \mathbb{Z}^+$,

- $f(a, b) \leq a + b$;
- $f(a, f(b, c)) = f(f(a, b), c)$; and
- Both $\binom{f(a,b)}{a}$ and $\binom{f(a,b)}{b}$ are odd numbers. (where $\binom{n}{k}$ denotes the binomial coefficients)

Ans. We claim the answer is binary OR (which clearly works), denoted by $\odot$. We'll denote $f(a, b)$ as $a \boxtimes b$, which is associative and surjective and prove that $a \boxtimes b$ is $a \odot b$ for $a, b \leq 2^k$ by induction on k . The base case $k = 0$ is clear (meaning $1 \boxtimes 1 = 1$).

Assume now it's true up to 2^{k-1} . First, note that by Lucas theorem, whenever $n < 2^k$ we get

$$n \boxtimes 2^k = 2^k \boxtimes n = 2^k + n.$$

(since the f -value is also $\leq 2^k + n$). Now, for $a, b < 2^{k-1}$ we get

$$(a + 2^{k-1}) \boxtimes b = 2^{k-1} \boxtimes a \boxtimes b = 2^{k-1} \boxtimes (a \odot b) = 2^{k-1} + a \odot b.$$

Similarly for the symmetric case. Finally,

$$\begin{aligned}(a + 2^{k-1}) \boxtimes (b + 2^{k-1}) &= a \boxtimes 2^{k-1} \boxtimes 2^{k-1} \boxtimes b = a \boxtimes 2^{k-1} \boxtimes b \\ &= (a + 2^{k-1}) \boxtimes b = 2^{k-1} + a \odot b.\end{aligned}$$

So all that remains is to show $2^k \boxtimes 2^k = 2^k$, which follows from surjectivity since $a \boxtimes b \geq \min\{a, b\}$. Thus induction done.

Problem 5. (RMM2011, P2, Day 1) Determine all positive integers n for which there exists a polynomial $f(x)$ with real coefficients, with the following properties:

- for each integer k , the number $f(k)$ is an integer if and only if k is not divisible by n ;
- the degree of f is less than n .

Ans. The only positive integers n which satisfy our conditions are the powers of prime numbers.

Applying the Lagrange's interpolation formula on some n values of f which are integral, one may conclude that $f \in \mathbb{Q}[X]$. Therefore, we may consider $F(X) = N \cdot f(X)$ where N is the common denominator of all the coefficients of f . Thus, $F \in \mathbb{Z}[X]$ with the condition that $N \mid F(k)$ if and only if $n \nmid k$. Let $d = \gcd(N, n)$. We see that if $n \nmid k$ then $d \mid N \mid F(k)$. Moreover, we claim that $d \nmid F(0)$. Indeed, choose $k = n$ and see that our condition gives that $N \nmid F(n)$. However, if $g = \frac{N}{d}$ then $\gcd(n, g) = 1$ and we have $n \equiv r \pmod{g}$ for some $1 \leq r \leq g - 1$. This yields that $F(n) \equiv F(r) \equiv 0 \pmod{g}$. Therefore, $d \nmid F(n)$, however, $F(n) \equiv F(0) \pmod{d}$. Now, we claim that $d = n$. Suppose not, and let $d < n$. Set $k = dm$ for some positive integer m relatively prime to n . We have $n \nmid dm$ and so $N \mid F(dm)$, in particular, $F(dm) \equiv F(0) \pmod{d}$, which is clearly false. We conclude that $d = n$. Thus, we have the new condition: $n \mid F(j)$ for all $1 \leq j \leq n - 1$ and $n \nmid F(0)$. Suppose that n is not a prime power. Then let p be a prime with exponent a such that $p^a \mid n$. We have $p^a \mid n \mid F(p^a)$ yielding that $p^a \mid F(0)$. Applying this for all prime powers dividing n , we get $n \mid F(0)$. Thus, we arrive at a contradiction. Note that the above argument doesn't yield a contradiction if n is a prime power since then $1 \leq p^a \leq n - 1$ does not hold.

For $n = p^a$, consider the polynomial $F(X) = \frac{(X+1) \cdots (X+p^a-1)}{p^b}$ where $b = v_p((p^a - 1)!)$. It can be seen that this indeed satisfies our hypothesis, by applying Kummer-Lucas theorem. Indeed, we see that $v_p(F(0)) = 0$ and $v_p(F(r)) = v_p\left(\binom{r+p^a-1}{r}\right) + v_p((p^a - 1)!) - v_p(p^b) = v_p\left(\binom{r+p^a-1}{r}\right) = a$ for all $1 \leq r \leq p^a - 1$.